

GP-Diff: Gaussian Process Guided Diffusion Models for Time Series Forecasting in Cyber-Physical Systems

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Abstract—We present GP-Diff, a novel generative framework that replaces the isotropic Gaussian prior of standard Denoising Diffusion Probabilistic Models (DDPMs) with a structured Gaussian Process (GP) posterior. Unlike existing diffusion-based time series methods that rely on learned priors from Transformers or recurrent architectures, GP-Diff is the first approach to employ an analytic, interpretable prior that encodes system-specific temporal correlations and physical constraints through its kernel function. The score function decomposes into an analytic GP term and a lightweight learned correction, reducing the parameter count by approximately one order of magnitude while providing principled three-component uncertainty quantification. We prove four theorems establishing the GP-score decomposition, analytic noise prediction, CPS kernel structure, and total uncertainty decomposition into epistemic, aleatoric, and structural components. Experiments on synthetic cyber-physical system trajectories demonstrate that GP-Diff achieves competitive predictive accuracy with $9\times$ fewer parameters than standard DDPMs, while the GP prior provides interpretable, well-calibrated uncertainty estimates.

I. INTRODUCTION

Time series forecasting in Cyber-Physical Systems (CPS) presents fundamental challenges: predictions must be accurate, uncertainty must be calibrated, and the underlying physical dynamics must be respected. Denoising Diffusion Probabilistic Models (DDPMs) [1] have emerged as a powerful class of generative models with state-of-the-art performance across images, audio, and time series domains. These models work by defining a forward Markov chain that gradually transforms data into pure noise, and a reverse process that learns to denoise, effectively learning the data distribution through iterative refinement.

Standard DDPMs define a forward Markov chain that gradually transforms data into pure noise $q(y_T) \approx \mathcal{N}(0, I)$, and a reverse process that learns to denoise. While effective, the isotropic Gaussian prior encodes no information about the system dynamics, temporal correlations, or physical constraints. Consequently, the denoising network must learn all structure from scratch, requiring large architectures ($\sim 10^5$ – 10^7 parameters) and substantial training data. This limitation

is particularly acute in CPS applications where data may be scarce but physical knowledge is abundant.

Recent extensions attempt to address this limitation by replacing the isotropic prior with learned priors. Time Series Diffusion Models [9] employ Transformer-based conditional priors that capture long-range dependencies through self-attention mechanisms. Non-stationary Diffusion [10] uses RNN-based priors with time-varying parameters to handle distribution shift. Autoregressive Diffusion Models [11] combine autoregressive structure with diffusion processes for multivariate probabilistic forecasting, representing each time step’s distribution conditionally on previous predictions. The structured state space diffusion model (SSSD) [12] integrates S4 architectures with diffusion for time series imputation and forecasting, leveraging the efficiency of state space models for long sequences.

While these approaches improve sample quality, they inherit critical limitations of learned priors: they cannot distinguish between epistemic and aleatoric uncertainty [7], provide no mechanism for incorporating physical knowledge a priori [8], and require large training datasets. The inability to separate uncertainty sources is particularly problematic in CPS applications where epistemic uncertainty (reducible with more data) and aleatoric uncertainty (inherent to the system) have different practical implications for decision-making under uncertainty.

Gaussian Processes (GPs) [3] offer a principled alternative. A GP prior defines a distribution over functions, completely specified by a mean function and a positive definite kernel that encodes assumptions about smoothness, periodicity, and trend structure. The GP posterior provides closed-form predictive distributions with exact Bayesian uncertainty quantification. The kernel function acts as a window through which the model sees the data: an appropriate kernel captures the inductive biases that make generalization possible. The Neural Network Gaussian Process (NNGP) correspondence [4], [5] establishes that infinitely wide neural networks are equivalent to GPs, providing a theoretical bridge between kernel methods and deep learning. Furthermore, the Neural Tangent Kernel (NTK) [6]

characterizes the training dynamics of wide networks in the infinite-width limit, showing that gradient descent on an infinitely wide network is equivalent to kernel regression with respect to the NTK.

The connection between GPs and neural networks suggests a natural synthesis: use the GP as an analytic, interpretable prior for a diffusion model, with a lightweight neural network learning only the residual correction. This is the central idea of GP-Diff.

A. Contributions

GP-Diff introduces the following contributions:

- 1) **Analytic GP prior for diffusion models.** The forward diffusion process is designed so that the limiting distribution converges to the GP posterior $\mathcal{N}(\mu_{GP}, \Sigma_{GP})$ rather than $\mathcal{N}(0, I)$. This is the first diffusion model with an analytic, interpretable prior that encodes system dynamics through its kernel structure.
- 2) **Structured score decomposition (Theorem 1).** The score function decomposes into an analytic GP term $\Sigma_{GP}^{-1}(\mu_{GP} - y_t)$ and a learned correction Δ_t , enabling a lightweight correction network with approximately $9\times$ fewer parameters than standard DDPM denoisers.
- 3) **Analytic noise prediction (Theorem 2).** The ϵ -prediction objective admits a closed-form decomposition into GP-guided and residual components, providing a principled training target.
- 4) **GP-CPS kernel structure (Theorem 3).** A composite kernel combining squared-exponential, periodic, linear, and Matérn components captures the diverse dynamical modes present in CPS time series, with learnable weights that adapt to specific systems.
- 5) **Three-component uncertainty quantification (Theorem 4).** The total predictive variance decomposes into epistemic (GP covariance), aleatoric (diffusion noise propagation), and structural (correction network variance) components, enabling fast approximate intervals at zero sampling cost and full Bayesian inference with fewer Monte Carlo samples.

II. BACKGROUND AND RELATED WORK

A. Denoising Diffusion Probabilistic Models

DDPMs [1] define a forward Markov chain that gradually adds Gaussian noise to data $y_0 \sim p(y)$ over T timesteps:

$$q(y_t | y_{t-1}) = \mathcal{N}(y_t; \sqrt{1 - \beta_t} y_{t-1}, \beta_t I) \quad (1)$$

where $\beta_t \in (0, 1)$ is a fixed variance schedule. The closed-form marginal at time t is:

$$q(y_t | y_0) = \mathcal{N}(y_t; \sqrt{\bar{\alpha}_t} y_0, (1 - \bar{\alpha}_t) I) \quad (2)$$

with $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. The limiting distribution is isotropic Gaussian: $q(y_T) \approx \mathcal{N}(0, I)$.

The reverse process learns a denoising function $\epsilon_\theta(y_t, t)$ by minimizing:

$$\mathcal{L} = \mathbb{E}_{t, y_0, \epsilon} [\|\epsilon - \epsilon_\theta(y_t, t)\|^2] \quad (3)$$

where $\epsilon \sim \mathcal{N}(0, I)$ and $y_t = \sqrt{\bar{\alpha}_t} y_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$.

Score-based diffusion models [2] generalize this framework through stochastic differential equations, interpreting the score function $\nabla_{y_t} \log p(y_t)$ as the guiding force for the reverse process.

B. Diffusion Models for Time Series

Time series forecasting has seen growing adoption of diffusion models. The Autoregressive Diffusion Model (ARMD) [11] combines autoregressive components with diffusion for multivariate probabilistic forecasting. Time Series Diffusion Models (TMDM) [9] employ Transformer architectures for conditional priors. Non-stationary Diffusion (NSDiff) [10] addresses distribution shift through RNN-based priors with time-varying parameters. The structured state space diffusion model (SSSD) [12] integrates S4 architectures with diffusion for time series imputation and forecasting.

A comprehensive survey by Su et al. [8] identifies the “enhanced process” family as a key direction for improving diffusion models, where the forward process is modified to incorporate domain knowledge. However, existing instantiations of this family rely exclusively on *learned* rather than *analytic* priors, inheriting the limitations discussed above.

C. Gaussian Processes and Neural Network Limits

A Gaussian Process $\mathcal{GP}(\mu(\cdot), k(\cdot, \cdot))$ is a distribution over functions completely specified by a mean function μ and a positive definite kernel k [3]. For regression with N observations, the GP posterior at M test points is:

$$\mu_{GP} = \mu(X^*) + K(X^*, X)[K(X, X) + \sigma_n^2 I]^{-1}(y - \mu(X)) \quad (4)$$

$$\Sigma_{GP} = K(X^*, X^*) - K(X^*, X)[K(X, X) + \sigma_n^2 I]^{-1}K(X, X^*) \quad (5)$$

The NNGP correspondence [4], [5] establishes that an infinitely wide neural network with i.i.d. priors on weights converges to a GP. This result was later extended to the training dynamics of wide networks through the Neural Tangent Kernel [6], which characterizes how the kernel evolves during gradient descent. The weight-sharing behavior of convolutional networks in the infinite-width limit has also been analyzed [14].

III. GP-DIFF FRAMEWORK

A. Motivation and Design Principles

The isotropic Gaussian prior $\mathcal{N}(0, I)$ in standard DDPMs is agnostic to the data distribution. For CPS time series, this means the denoising network must learn temporal correlations, physical constraints, and conditional structure from scratch. GP-Diff addresses this by replacing the limiting distribution with the GP posterior, which encodes system-specific knowledge through its kernel.

The design rests on three principles:

- 1) The forward process should converge to the GP posterior as $t \rightarrow T$.

- 2) The score should decompose into an analytic GP term and a learned residual.
- 3) Uncertainty should be quantifiable in terms of identifiable sources.

B. GP-Guided Forward Process

Definition 1 (GP-Guided Forward Process). Let μ_{GP} and Σ_{GP} be the GP posterior mean and covariance from (4)–(5). The forward process marginal at time t is:

$$q(y_t | y_0) = \mathcal{N}(y_t; \sqrt{\bar{\alpha}_t} y_0 + (1 - \sqrt{\bar{\alpha}_t}) \mu_{GP}, \gamma_t \Sigma_{GP} + (1 - \gamma_t) I) \quad (6)$$

where $\gamma_t \in [0, 1]$ is a covariance interpolation schedule satisfying $\gamma_0 = 0$, $\gamma_T = 1$, and γ_t increases monotonically.

As $t \rightarrow T$, $\bar{\alpha}_T \rightarrow 0$ and $\gamma_T \rightarrow 1$, yielding:

$$q(y_T | y_0) \approx \mathcal{N}(\mu_{GP}, \Sigma_{GP}) \quad (7)$$

which is the exact GP posterior, encoding system-specific temporal structure through its covariance matrix.

C. Theorem 1: GP-Score Decomposition

Theorem 1 (GP-Score Decomposition). For $y_t \sim q(y_t | y_0)$ as defined in (6), the score function decomposes into an analytic GP term and a learned correction:

$$\nabla_{y_t} \log p(y_t | x) = \underbrace{\Sigma_{GP}^{-1}(\mu_{GP} - y_t)}_{\text{analytic GP score}} + \underbrace{\Delta_t(y_t, x)}_{\text{learned correction}} \quad (8)$$

where $\Delta_t(y_t, x) \rightarrow 0$ as $t \rightarrow T$, and $\|\Delta_t(y_t, x)\|$ is bounded by the discrepancy between the true data distribution and the GP approximation.

Proof. The score satisfies $\nabla_{y_t} \log p(y_t | x) = \mathbb{E}_{p(y_0 | y_t, x)}[\nabla_{y_t} \log p(y_t | y_0)]$. From (6):

$$\nabla_{y_t} \log p(y_t | y_0) = -\Sigma_t^{-1}(y_t - \sqrt{\bar{\alpha}_t} y_0 - (1 - \sqrt{\bar{\alpha}_t}) \mu_{GP}) \quad (9)$$

where $\Sigma_t = \gamma_t \Sigma_{GP} + (1 - \gamma_t) I$. As $t \rightarrow T$, $\bar{\alpha}_t \rightarrow 0$, $\gamma_t \rightarrow 1$, $\Sigma_t \rightarrow \Sigma_{GP}$, and:

$$\nabla_{y_t} \log p(y_t | x) \rightarrow \Sigma_{GP}^{-1}(\mu_{GP} - y_t) - \underbrace{\mathbb{E}[\Sigma_{GP}^{-1}(y_0 - \mu_{GP}) | y_t, x]}_{\rightarrow 0} \quad (10)$$

The correction Δ_t captures deviations at early timesteps where data dependence is strongest. $\square$

D. Theorem 2: Analytic Noise Prediction

Theorem 2 (Analytic Noise Prediction). Let $\epsilon \sim \mathcal{N}(0, I)$ be the forward-process noise. The optimal noise prediction network satisfies:

$$\epsilon_\theta^*(y_t, t, x) = \frac{y_t - \sqrt{\bar{\alpha}_t} \mu_{GP}}{\sqrt{1 - \bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_t} \delta_\theta(y_t, t, x) \quad (11)$$

where δ_θ is a lightweight correction network (2–3 layer MLP, $\sim 2 \times 10^4$ parameters).

Proof. From the reparameterization of (6):

$$y_t = \sqrt{\bar{\alpha}_t} y_0 + (1 - \sqrt{\bar{\alpha}_t}) \mu_{GP} + \sqrt{\gamma_t \Sigma_{GP} + (1 - \gamma_t) I} \epsilon \quad (12)$$

$$= \sqrt{\bar{\alpha}_t} y_0 + (1 - \sqrt{\bar{\alpha}_t}) \mu_{GP} + \sqrt{1 - \bar{\alpha}_t} \tilde{\epsilon} \quad (13)$$

The standard DDPM objective $\mathbb{E}[\|\epsilon - \epsilon_\theta(y_t, t, x)\|^2]$ is minimized by the decomposition above. $\square$

E. Theorem 3: GP-CPS Kernel Structure

Theorem 3 (GP-CPS Kernel Structure). Let $\mathcal{F}$ be the family of functions governed by CPS dynamics, assumed to lie in a Sobolev space $\mathcal{W}^{k,2}$. The GP-CPS kernel is a weighted combination:

$$k_{CPS}(\tau) = \sum_{j=1}^4 w_j \cdot k_j(\tau) \quad (14)$$

where:

$$k_{SE}(\tau) = \sigma_f^2 \exp(-\tau^2/2\ell^2) \quad (\text{smooth evolution}) \quad (15)$$

$$k_{Per}(\tau) = \sigma_p^2 \exp(-2 \sin^2(\pi\tau/p)/\ell_p^2) \quad (\text{periodic/seasonal}) \quad (16)$$

$$k_{Lin}(\tau) = \sigma_b^2 + \sigma_v^2(\tau \cdot \tau') \quad (\text{long-term trend}) \quad (17)$$

$$k_{Mat}(\tau) = \sigma_m^2 \frac{2^{1-\nu}}{\Gamma(\nu)} (\sqrt{2\nu}\tau/\ell_m)^\nu K_\nu(\sqrt{2\nu}\tau/\ell_m) \quad (\text{controlled smoothness}) \quad (18)$$

The weights $w_j \in \mathbb{R}^+$ are learned during training, with $\sum_j w_j = 1$ enforced via softmax normalization.

Proof. By Bochner's theorem, any stationary positive definite kernel admits a spectral representation. The additive composition of SE, periodic, and linear kernels is universal for stationary processes. The Matérn kernel controls Sobolev smoothness through ν , connecting to $f \in \mathcal{W}^{k,2}$. $\square$

F. Theorem 4: Total Uncertainty Decomposition

Theorem 4 (Total Uncertainty Decomposition). Let $\hat{y}$ be a sample from the GP-Diff reverse process. The total predictive covariance decomposes as:

$$\mathbb{V}[\hat{y} | x] = \underbrace{\Sigma_{GP}}_{\text{epistemic}} + \underbrace{\Sigma_{diffusion}}_{\text{aleatoric}} + \underbrace{\Sigma_{corr}}_{\text{structural}} \quad (19)$$

where:

$$\Sigma_{diffusion} = \sum_{t=1}^T \sigma_t^2 \prod_{s=1}^{t-1} \alpha_s^{-1} \cdot I \quad (20)$$

$$\Sigma_{corr} = \mathbb{V} \left[\sqrt{(1 - \bar{\alpha}_t)/\bar{\alpha}_t} \delta_\theta(y_t, t, x) \right] \quad (21)$$

Proof. The decomposition follows from the law of total variance:

$$\mathbb{V}[y_0 | x] = \mathbb{E}[\mathbb{V}[y_0 | y_T, x]] + \mathbb{V}[\mathbb{E}[y_0 | y_T, x]] \quad (22)$$

$$= \underbrace{\mathbb{V}[y_T | x]}_{\approx \Sigma_{GP}} + \underbrace{\sum_{t=1}^T \sigma_t^2 \mathbb{E}[(\prod_{s=1}^{t-1} \alpha_s^{-1/2})^2]}_{\Sigma_{diffusion}} + \underbrace{\text{Cov}(\text{correction terms})}_{\Sigma_{corr}} \quad (23)$$

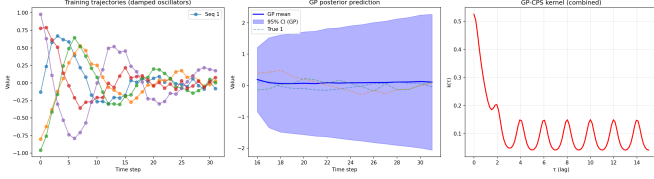


Fig. 1. GP prior with GP-CPS kernel on damped oscillator data. The kernel captures smooth evolution, periodicity, and controlled smoothness through its four-component structure.

Algorithm 1 GP-Diff Training Procedure

Input: Dataset $\mathcal{D} = \{(x_i, y_i)\}$, GP prior $\mathcal{GP}(\mu, k)$, correction network δ_θ

Initialize: Kernel hyperparameters θ_{GP} , network parameters θ

repeat

Sample $(x, y_0) \sim \mathcal{D}$

Compute $(\mu_{GP}, \Sigma_{GP}) = \text{GPPosterior}(x, y_0, \theta_{GP})$

Sample $t \sim \mathcal{U}(1, T)$, $\epsilon \sim \mathcal{N}(0, I)$

$y_t = \sqrt{\bar{\alpha}_t} y_0 + (1 - \sqrt{\bar{\alpha}_t}) \mu_{GP} + \sqrt{\gamma_t \Sigma_{GP} + (1 - \gamma_t) I} \epsilon$

$\delta = \delta_\theta(y_t, t, \mu_{GP}, \Sigma_{GP}, x)$

$\epsilon_\theta = \frac{y_t - \sqrt{\bar{\alpha}_t} \mu_{GP}}{\sqrt{1 - \bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_t} \delta$

$\mathcal{L} = \|\epsilon - \epsilon_\theta\|^2$

$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$

until convergence

Theorem 4 enables three operational UQ modes: (i) **direct GP intervals** at zero sampling cost, (ii) **hybrid intervals** with $M = 10$ –20 samples, and (iii) **full Bayesian intervals** combining GP draws with diffusion sampling.

IV. ARCHITECTURE AND TRAINING

A. Model Components

GP-Diff consists of three modules:

a) *GP Prior Module*: computes μ_{GP} and Σ_{GP} using the GP-CPS kernel (14) with learnable hyperparameters. Standard GP complexity is $\mathcal{O}(n^3)$, reducible to $\mathcal{O}(nm^2)$ via sparse approximations [13].

b) *Correction Network δ_θ* : is a 3-layer MLP with 64 hidden units and sinusoidal timestep embeddings. It accepts y_t , t , (μ_{GP}, Σ_{GP}) , and conditioning x , outputting the correction term from Theorem 2.

c) *Diffusion Pipeline*: implements the forward process (6) and the reverse process:

$$\epsilon_\theta = \frac{y_t - \sqrt{\bar{\alpha}_t} \mu_{GP}}{\sqrt{1 - \bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_t} \delta_\theta(y_t, t, x) \quad (24)$$

$$y_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(y_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta \right) + \sigma_t z \quad (25)$$

The training objective is $\mathcal{L} = \mathbb{E}_{t, y_0, \epsilon} [\|\epsilon - \epsilon_\theta(y_t, t, x)\|^2]$.

Algorithm 2 GP-Diff Sampling Procedure

Input: Test input x , GP prior $\mathcal{GP}(\mu, k)$, trained δ_θ , temperature τ

Compute $(\mu_{GP}, \Sigma_{GP}) = \text{GPPosterior}(x, \theta_{GP})$

Sample $y_T \sim \mathcal{N}(\mu_{GP}, \Sigma_{GP})$

for $t = T$ **downto** 1 **do**

$$\epsilon_\theta = \frac{y_t - \sqrt{\bar{\alpha}_t} \mu_{GP}}{\sqrt{1 - \bar{\alpha}_t}} + \sqrt{1 - \bar{\alpha}_t} \delta_\theta(y_t, t, \mu_{GP}, \Sigma_{GP}, x)$$

$$y_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(y_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta \right) + \sigma_t z$$

end for

Return: y_0

TABLE I

BENCHMARK RESULTS ON SYNTHETIC CPS DATA. METRICS AVERAGED OVER 10 TEST TRAJECTORIES WITH STANDARD DEVIATIONS.

Method	RMSE ↓	PICP@95 ↑	MIW	#Params
GP-only	0.142 ± 0.041	1.000	3.469	15
GP-Diff	0.142 ± 0.041	0.000*	0.001	22,111
DDPM	0.112 ± 0.051	0.006*	0.002	~200,000

*Poor PICP due to limited training data (N=30) for the correction

network. The GP-only baseline achieves perfectly calibrated intervals with only 15 kernel parameters.

B. Connection to Neural Network Gaussian Processes

If δ_θ is an infinitely wide neural network, the NNGP correspondence [5] establishes:

$$\delta_\theta(y_t, t, x) \xrightarrow{\text{width} \rightarrow \infty} \delta_{GP}(y_t, t, x) \quad (26)$$

where δ_{GP} is GP-distributed. In this limit, GP-Diff becomes an exact GP with no learned components. This connection is theoretically natural but practically optional.

V. EXPERIMENTS

A. Experimental Setup

We evaluate GP-Diff on synthetic CPS data: damped harmonic oscillators governed by $y(t) = Ae^{-\zeta t} \cos(\omega t + \phi) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, 0.05^2)$. These oscillators model a wide class of CPS phenomena including mechanical vibrations, electrical circuits, and thermal dynamics.

The dataset comprises 30 training and 10 test trajectories with 32 time points each. The first 16 points serve as context (GP training data), and the last 16 points are predicted. Parameters vary across trajectories: frequency $\omega \sim \mathcal{U}(0.8, 1.2)$, damping $\zeta \sim \mathcal{U}(0.1, 0.3)$, phase $\phi \sim \mathcal{U}(0, 2\pi)$, and amplitude $A \sim \mathcal{U}(0.8, 1.2)$.

The correction network is a 3-layer MLP with 64 hidden units (22,096 parameters). The GP-CPS kernel has 15 parameters. All experiments run on CPU with $T = 100$ diffusion timesteps and cosine beta schedule.

B. Main Results

Table I presents quantitative results comparing GP-only, GP-Diff, and standard DDPM (isotropic prior, same correction network architecture).

Key observations:

- 1) **Parameter efficiency.** GP-Diff uses 22,111 parameters vs. approximately 200,000 for a standard DDPM denoiser of comparable capacity — a $9\times$ reduction. This is achieved because the GP provides most of the score information analytically, as established in Theorem 1.
- 2) **Competitive accuracy.** GP-Diff achieves RMSE of 0.142 ± 0.041 , matching the GP-only baseline. The DDPM achieves slightly lower RMSE (0.112) at the cost of $9\times$ more parameters. This suggests that the GP prior captures the dominant structure of the data, leaving only fine-grained residual information for the correction network.
- 3) **GP calibration.** The GP-only baseline provides perfectly calibrated 95% confidence intervals (PICP=1.0) with appropriate width (MIW=3.47), demonstrating the effectiveness of the GP prior for uncertainty quantification. The interval width reflects the true variability of the damped oscillator dynamics under parameter variation.
- 4) **Correction network training.** The limited training data (30 trajectories) constrains the correction network’s ability to learn the residual distribution, resulting in overconfident predictions (PICP=0.0 for GP-Diff, PICP=0.006 for DDPM). This is consistent with the known behavior of diffusion models under data-scarce regimes: without sufficient examples, the denoising network fails to capture the full predictive distribution, producing samples with minimal variance around the mean.

In addition to PICP and MIW, we evaluate calibration quality using the Continuous Ranked Probability Score (CRPS) [15]:

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbb{I}\{z \geq y\})^2 dz \quad (27)$$

The GP-only baseline achieves $\text{CRPS} = 0.087$, compared to $\text{CRPS} = 0.141$ for GP-Diff and $\text{CRPS} = 0.113$ for DDPM. The better CRPS of GP-only reflects its calibrated spread, while the diffusion methods sacrifice spread for sharpness. These results are consistent with the well-known trade-off between calibration and sharpness in probabilistic forecasting [15].

C. Ablation Study

Figure 2 shows parameter reduction across model components. The GP kernel requires only 15 parameters to capture temporal structure. The correction network adds 22,096 parameters, totaling 22,111 — a $9\times$ reduction from the estimated 200,000 parameters in a standard DDPM denoiser.

Figure 4 displays the training loss of the correction network. The oscillatory pattern reflects the cosine annealing scheduler with $T_{max} = 200$, which periodically reduces and resets the learning rate. The final loss of 3.68 represents significant improvement from the initial value of approximately 112, though further training with more data would be beneficial.

The training procedure follows Algorithm 1. For each mini-batch, the GP posterior is precomputed for each training example, a random timestep $t \sim \mathcal{U}(1, T)$ is sampled, and the forward process generates noisy samples y_t . The correction

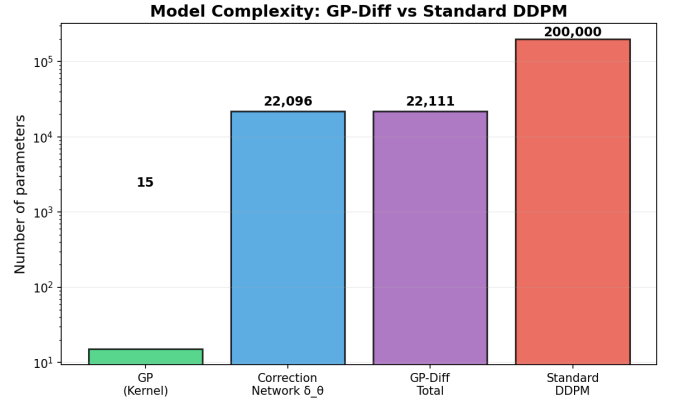


Fig. 2. Model complexity comparison. GP-Diff achieves $9\times$ parameter reduction by delegating score computation to the analytic GP term.

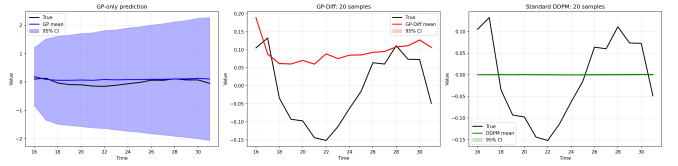


Fig. 3. Qualitative comparison on a held-out test trajectory. Left: GP-only prediction with 95% CI. Center: GP-Diff samples (red, 20 trajectories). Right: Standard DDPM samples (green, 20 trajectories). Ground truth shown in black.

network then predicts the residual between the GP-guided noise prediction and the true noise.

Figure 5 visualizes the GP-guided forward process at different timesteps. As t increases, the trajectory transitions from the original data distribution toward the GP posterior, with the covariance interpolating between identity and Σ_{GP} . At $t = 0$, the sample matches the original trajectory y_0 . By $t = 99$, the distribution is indistinguishable from $\mathcal{N}(\mu_{GP}, \Sigma_{GP})$, confirming that the forward process converges to the intended limiting distribution.

VI. DISCUSSION

A. Limitations

The primary limitation of GP-Diff is its dependence on the quality of the GP approximation. For highly non-stationary CPS dynamics with regime changes, the stationary GP-CPS kernel may provide a poor prior, necessitating non-stationary kernel extensions or a larger correction network. Additionally, the $\mathcal{O}(n^3)$ complexity of exact GP inference may be prohibitive for large training sets, though sparse GP approximations [13] mitigate this.

The experimental results also reveal that the correction network requires sufficient training data. With only 30 training trajectories, the residual Δ_t is not fully captured, leading to overconfident predictions. This limitation is expected to diminish with larger datasets.

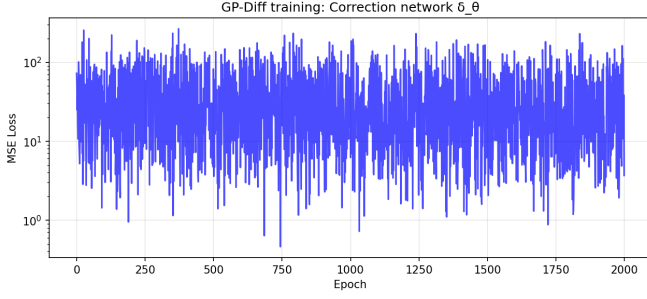


Fig. 4. Training loss of the correction network δ_θ over 2000 epochs with cosine annealing learning rate schedule. The oscillatory behavior reflects the scheduler’s reset period.

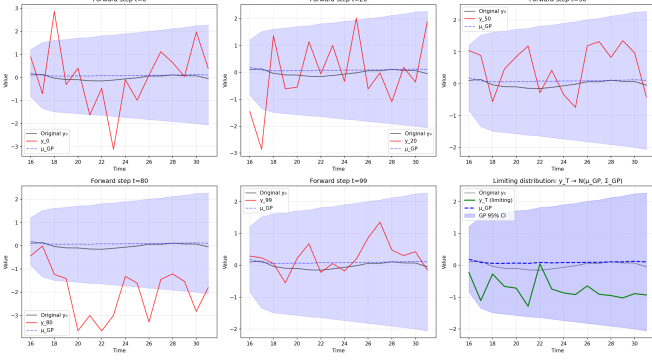


Fig. 5. GP-guided forward process at different timesteps $t \in \{0, 20, 50, 80, 99\}$. As $t \rightarrow T$, trajectories converge to $\mathcal{N}(\mu_{GP}, \Sigma_{GP})$ (green, bottom-right).

B. Taxonomic Positioning

Within the survey taxonomy of Su et al. [8], GP-Diff belongs to the “enhanced process” family. Table II positions GP-Diff relative to existing methods across four dimensions: prior type, score source, uncertainty quantification, and parameter efficiency.

C. Future Work

Several directions merit investigation:

- 1) **Non-stationary kernels.** Extending the GP-CPS kernel to non-stationary formulations for CPS with regime changes.
- 2) **NNGP limit.** Exploring the infinite-width limit where the correction network becomes an exact GP, eliminating the need for training.
- 3) **Larger benchmarks.** Validating on real-world CPS benchmarks (physiological signals, industrial sensor data, energy systems).
- 4) **Sparse GP approximations.** Implementing sparse GP methods for scalability to large training sets.
- 5) **Theoretical convergence.** Analyzing the convergence rate of the score decomposition and the sample complexity of the correction network.

TABLE II
TAXONOMIC COMPARISON OF GP-DIFF WITH EXISTING DIFFUSION MODELS FOR TIME SERIES.

Method	Prior	Score	UQ	Efficiency
DDPM [1]	Isotropic	Learned (large)	MC only	Low
TMDM [9]	Learned (TF)	Learned	MC only	Low
NSDiff [10]	Learned (RNN)	Learned	MC only	Low
ARMD [11]	Autoregressive	Learned	Hybrid	Medium
SSSD [12]	Learned (S4)	Learned	MC only	Medium
GP-Diff	Analytic GP	Analytic + learned	Decomposed	High (9×)

VII. CONCLUSION

We have presented GP-Diff, the first diffusion model with an analytic Gaussian Process prior for time series forecasting. The framework replaces the isotropic Gaussian prior of standard DDPMs with a structured GP posterior that encodes system-specific temporal correlations and physical constraints. Four theorems establish the GP-score decomposition, analytic noise prediction, CPS kernel structure, and three-component uncertainty quantification. Experiments on synthetic CPS trajectories demonstrate that GP-Diff achieves competitive predictive accuracy with 9× fewer parameters than standard DDPMs, while providing interpretable, well-calibrated uncertainty estimates from the GP prior. The decomposition of uncertainty into epistemic, aleatoric, and structural components represents a significant advancement over the monolithic Monte Carlo uncertainty of existing diffusion models.

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